

PROFESSIONAL SUMMARY

I am a final-year Mathematics undergraduate at Universitas Diponegoro with a 3.92/4.00 GPA. My primary interests are pure mathematics, especially real and complex analysis, graph theory, abstract algebra, topology, and combinatorics. I also have experience in university mathematics teaching, olympiad coaching, academic-content development, data analysis, and computational projects.

EDUCATION

Sep 2023 — Present ONGOING	Universitas Diponegoro B.Sc. in Mathematics · GPA 3.92 / 4.00 Chair, UKM KOMIPA UNDIP (2025); teaching and tutorial experience in Real Analysis II; active in the Mathematics Student Association; ON-MIPA delegate and mathematics competition awardee.
Jun 2020 — Mar 2023 COMPLETED	SMA Negeri 1 Sulang Mathematics and Natural Sciences · Final Grade 86.55 / 100.00 Earned regional and national mathematics awards, represented the school in OSN Informatics, and provided peer mathematics mentoring.

RESEARCH & ACADEMIC WORKS

2026 DRAFT	Distance Spectra of Corona Graphs of Complete Graphs and Paths via Invariant Subspaces Desca Affajry Rais · Suryoto, S.Si., M.Si. Studies the distance spectrum of $K_n \odot P_m$ using block matrices, Kronecker products, and invariant subspaces.
2026 DRAFT	Connectivity of Int-Graphic Topology on Directed Graphs Desca Affajry Rais · YD Sumanto · Solikhin Characterizes connectivity of finite int-graphic Alexandroff spaces through minimal open neighborhoods and the out-neighborhood inclusion preorder.
2026 PRESENTED	Modular Total Edge Irregular Labeling of Corona Product of Cycle C_4 with Path and Null Graphs The 16th ISNPINSA · Oral Presentation Developed explicit labeling constructions and established exact modular total edge irregularity strength results.
2026 COMPLETED	Inequality Problem Solving Book Desca Affajry Rais · Drs. YD Sumanto, M.Si. · First Edition · 174 pages Proof-oriented mathematics book covering inequality theory, rigorous proofs, worked examples, and exercises. The manuscript is complete; Indonesian Intellectual Property (HKI) registration is planned.

PROFESSIONAL EXPERIENCE

Aug 2026 — Present ONGOING	University Mathematics Tutor Tutorkuliah.id Tutoring in advanced undergraduate mathematics with emphasis on conceptual understanding and rigorous problem solving.
Jan 2026 — Present ONGOING	Mathematics Olympiad Coach KOMIPA UNDIP University-level training in Real Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, and Combinatorics.
Apr 2026 — Jun 2026 COMPLETED	Teaching Assistant Asosiasi Matematika Terapan UNDIP Real Analysis tutorials, proof-oriented problem solving, and academic consultation.
Dec 2025 — Mar 2026 COMPLETED	Mathematics Item Writer SainsIn Authored quantitative-reasoning items, detailed solutions, instructional videos, and quality-control materials.

Nov 2025 — Jan 2026 COMPLETED	Data Analyst Emergencyy Call Data cleaning, SQL querying, Excel reporting, KPI dashboards, visualization, and workflow documentation.
Nov 2025 — Dec 2025 COMPLETED	Mathematics Item Writer SNBT.ID Authored high-difficulty Quantitative Knowledge and Mathematical Reasoning items with structured solutions.
May 2025 COMPLETED	Project-Based Virtual Intern · Data Scientist ID/X Partners × Rakamin Academy Built a credit-risk prediction pipeline using Logistic Regression and Random Forest.
Mar 2025 COMPLETED	Project-Based Virtual Intern · Big Data Analyst PT Kimia Farma Tbk × Rakamin Academy Built a BigQuery analytical table and a Looker Studio performance dashboard.

LEADERSHIP & ORGANIZATIONS

Jan 2025 — Dec 2025 COMPLETED	Chairman Mathematics and Sciences Olympiad Club (KOMIPA UNDIP) Led strategy, daily operations, olympiad mentoring, ON-MIPA delegation, and external representation.
Jan 2025 — Dec 2025 COMPLETED	Senior Staff, Education and Reasoning Department Mathematics Student Association, Universitas Diponegoro Delivered mathematics tutorials, educational content, and learning-material development.
Feb 2024 — Jan 2025 COMPLETED	Junior Staff, Education and Reasoning Department Mathematics Student Association, Universitas Diponegoro Coordinated Academic Delegation, mentoring for 100+ students, and received Best Junior Staff and Best Department Staff recognition.
Oct 2023 — Dec 2024 COMPLETED	Staff, Networking and Public Relations Department KOMIPA UNDIP Supported ON-MIPA selection, served as Mathematics Liaison Officer, and led UKM Roadshow 2024.
2020 — 2023 COMPLETED	Moderator, Mentor, and Board of Governor Brainly Progressed through multiple moderation and mentoring roles, including supervision of 29 apprentice moderators.
2020 — 2022 COMPLETED	Organizational Leadership Pelajar Islam Indonesia (PII), Rembang Served as Head of Organizational Development and later General Secretary of the Rembang Regional Board.

COMMITTEE & VOLUNTEER EXPERIENCE

Facilitator Division Coordinator / Koordinator Divisi Fasilitator 27th Undip's Mathematics Competition · Jan 2025 — Feb 2026	Curriculum Division Staff / Staf Divisi Kurikulum LKMM-TD FSM 2025 · Jan — Oct 2025
Event Division Staff / Staf Divisi Acara Introduction, Gathering, & Adaptation Mathematics 2025 · Mar — Oct 2025	Event Division Staff / Staf Divisi Acara Industrial Visit Mathematics 2025 · Mar — Aug 2025
Equipment Division Staff / Staf Divisi Perlengkapan Math Action for Social and Education 2025 · Mar — Jun 2025	Public Relations Division Staff / Staf Divisi Humas Leadership Training FSM 2025 · Mar — Apr 2025
Event Division Coordinator / Koordinator Divisi Acara 26th Undip's Mathematics Competition · Mar 2024 — Feb 2025	Chief Executive / Ketua Pelaksana UKM Roadshow 2024 · Jun — Sep 2024
Volunteer Teacher / Pengajar Sukarela Mathematics Clinic 2024 · Aug 2024	Equipment Division Coordinator / Koordinator Divisi Perlengkapan ON-MIPA UNDIP Selection 2024 · Oct 2023 — Feb 2024
Mathematics Liaison Officer / Liaison Officer Matematika ON-MIPA Coaching Center 2024 · Feb 2024	

SELECTED PROJECTS

Feb 2026 COMPLETED	Crowd Optimization and Smart Routing Strategy at Dufan Ancol Operations Research · Mathematical Modeling Applied M/M/s queueing, Little’s Law, greedy optimization, and knapsack concepts; proposed 14,000 daily visitors, a <45-minute waiting-time target, and 75% maximum utilization.
Feb 2026 ONGOING	DMath Learning Mathematics Education · EdTech Ongoing mathematics learning platform for structured materials, guided practice, problem banks, olympiad preparation, and learning tools.
Feb 2026 COMPLETED	Bike Sharing Analytics Python · EDA · Streamlit Identified weekday demand peaks, weekend casual-user behavior, and environmental drivers through end-to-end exploratory analysis.
Dec 2025 COMPLETED	Financial Anomaly Detection System Scikit-Learn · K-Means · Random Forest Combined unsupervised pattern discovery with supervised transaction classification.
May 2025 COMPLETED	Credit Risk Prediction Analysis SMOTE · Logistic Regression · Random Forest Analyzed 466,285 observations and reported Random Forest performance of 99.70% accuracy, 99.94% precision, and 99.46% recall.
Mar 2025 COMPLETED	Kimia Farma Business Performance Analysis BigQuery · SQL · Looker Studio Built a centralized analytical table and multidimensional performance dashboard.
Mar–Apr 2025 COMPLETED	Investment Portfolio Optimization Markowitz · Efficient Frontier · Sharpe Ratio Analyzed portfolio allocation across KLBFIJK, HEALJK, and KAEFIJK using risk-return optimization.

SCHOLARSHIPS & TRAINING

Nov 2024 — Present ONGOING	BSI Scholarship Inspirasi Batch 4 Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities. Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.
Feb — Jul 2026 COMPLETED	Coding Camp 2026 — Data Scientist Coding Camp powered by DBS Foundation Coding Camp powered by DBS Foundation
Mar — Jun 2026 COMPLETED	Microsoft Elevate Training Center Data Science with Microsoft Fabric · Generative AI with Microsoft Azure Data Science with Microsoft Fabric · Generative AI with Microsoft Azure
Jan — Mar 2026 COMPLETED	IDCamp 2025 — Intermediate Data Scientist Indosat Ooredoo Hutchison Indosat Ooredoo Hutchison
Oct — Dec 2025 COMPLETED	IDCamp 2025 — Beginner Data Scientist Indosat Ooredoo Hutchison Indosat Ooredoo Hutchison

CERTIFICATIONS

UNDIP SEU English Proficiency Test EPT 65 · TOEFL Prediction equivalent 549 · 17 Sep 2026	TOEFL ITP Prediction Test 590 / 677
AI Engineer Bootcamp / Bootcamp AI Engineer	Foundations of Artificial Intelligence / Belajar Dasar AI
Foundations of SQL / Belajar Dasar SQL	Computer Networking Fundamentals / Fundamental Jaringan Komputer

Data Analyst — Basic Level ITBOX · Mar 2025	Python Programming Essentials: A Beginner's Guide Rakamin Academy · Mar 2025
Classical Cryptography for Beginners Cyber Academy Indonesia · Jul 2024	Introduction to Information Security Cyber Academy Indonesia · Jul 2024
Student Leadership and Management Training — FSM 2024 BEM FSM UNDIP · Apr 2024	Introduction to Data Science with Python DQLab · Jan 2024
Python Fundamentals for Data Science DQLab · Jan 2024	Basic Student Management Skills Training / LKMM Pra Dasar Sep 2023
Python Programming for Beginners CODEPOLITAN · Credential Z7R8TDD · Jul 2024 — Jul 2027	

SKILLS

Mathematics & Theory Pure Mathematics · Real Analysis · Complex Analysis · Graph Theory · Abstract Algebra · Topology · Combinatorics · Mathematical Modeling · Formal Proof Writing	Technical & Computing Python · SQL · LaTeX · Scikit-Learn · Looker Studio · Microsoft Excel · Microsoft Fabric · Microsoft Azure
Data & Analytics Data Analysis · Exploratory Data Analysis · Machine Learning · Data Visualization · Statistical Analysis · Data Engineering	Academic & Teaching Research · Academic Writing · Mathematics Olympiad · Teaching · Curriculum Development · Assessment Design · Problem Solving

AWARDS & ACHIEVEMENTS

Silver Medalist / Medali Perak · Isaac Newton Competition 2026 · OLIMNUS × UIN Sunan Kalijaga Jul 2026	Gold Award · KTOM December 2025 Dec 2025
Silver Medalist / Medali Perak · Future Science Olympiad · POSI Jun 2026	Special Honour Participant · International Youth Math Challenge Dec 2025
Bronze Medalist / Medali Perunggu · Mathematical Analysis and Geometry Day XVII · Institut Teknologi Bandung May 2026	1st Place / Juara 1 · LOSEMA X Nov 2025
Gold Medalist / Medali Emas · Cendekia Natural Science Competition Apr 2026	1st Place / Juara 1 · Olimnus Science 2025 Nov 2025
1st Place / Juara 1 · ON-MIPA UNDIP Selection 2026 Apr 2026	Silver Medalist / Medali Perak · Isaac Newton Competition 2025 Jul 2025
1st Place / Juara 1 · Indonesia Science Competition (KSI) Mar 2026	3rd Place / Juara 3 · ASOE 4.0 May 2025
Gold Award · KTOM March 2026 Mar 2026	4th Place · ON-MIPA UNDIP Selection 2025 Mar 2025
1st Place / Juara 1 · Indonesia Science Competition (InSC) Feb 2026	Finalist / Finalis · MATHLICIOUS 2022 Nov 2022
4th Place · Honorable Mention 1 · MATICS Mathematics Olympiad · Hasanuddin University Feb 2026	1st Place / Juara 1 · ALHAMBRA Mathematics Competition Feb 2022
Silver Award · KTOM February 2026 Feb 2026	Bronze Medalist / Medali Perunggu · Insight Arjuna Competition Oct 2021
Bronze Medalist / Medali Perunggu · POSI Science Competition Jan 2026	Gold Medalist / Medali Emas · Dirgantara Science Competition Oct 2021
2nd Place / Juara 2 · Indonesian Youth Science Olympiad #6 Jan 2026	Silver Medalist / Medali Perak · Indonesian Youth Science Competition Oct 2021
2nd Place / Juara 2 · National Science League #2 Jan 2026	Bronze Medalist / Medali Perunggu · Bimage Competition Season 2 Sep 2021
Silver Medalist / Medali Perak · Einstein Science Competition 2026 Jan 2026	Silver Medalist / Medali Perak · POSI Science Competition Sep 2021
	Top 10 · National Calculus Olympiad Nusantara Aug 2021